

Fructosamine

Test Details

METHODOLOGY

Colorimetric

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

Evaluate diabetic control, reflecting diabetic control over a shorter time period (one to three weeks) than that represented by hemoglobin A_{1c} (four to eight weeks). Indicated as an index of longer-term control than glucose levels, especially in diabetic subjects with abnormal hemoglobins and in type 1 diabetes in children.¹

Custom Additional Information

Fructosamine is found in the plasma of both normal and diabetic individuals. “Fructosamine” is the term used to describe proteins that have been glycated (ie, are derivatives of the nonenzymatic reaction product of glucose and albumin). It has been advocated as an alternative test to hemoglobin A_{1c} for the monitoring of long-term diabetic control.

Fructosamine and hemoglobin A_{1c} do not measure exactly the same thing, since fructosamine has a shorter half-life and appears to be more sensitive to short-term variations in glucose levels; however, this is not necessarily a disadvantage. Fructosamine is clearly superior in patients with abnormal hemoglobins because of the interference of abnormal hemoglobins in the anion-exchange chromatography methods for Hb A_{1c}. Published reference interval for apparently healthy subjects between age 20 and 60 is 205–285 µmol/L and in a poorly-controlled diabetic population is 228–563 µmol/L with a mean of 396 µmol/L.

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.2 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Gel-barrier tube, transport tube, green-top (heparin) tube, **or** lavender-top (EDTA) tube

Collection Instructions

Separate serum or plasma from cells within 45 minutes of collection.

Stability Requirements

Temperature	Period
Room temperature	7 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x2

Storage Instructions

Maintain specimen at room temperature.

Causes for Rejection

Gross hemolysis; improper labeling; gross lipemia

References

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Footnotes

1. Cefalu WT, Mejia E, Puente GR, Fleishhacker D, Macaulay K. Correlation of serum fructosamine activity in type I diabetic children. *Am J Med Sci*. 1989 Apr; 297(4):244-246. PubMed 2705464

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